

TABLE V: The renormalized decay amplitude ratio $\Phi_{B_s}^{\text{ren}}/\Phi_{B_l}^{\text{ren}}$ for both the APE- and HYP-smeared data sets. Errors shown are statistical only.

am_l/am_h	$\Phi_{B_s}^{\text{ren}}/\Phi_{B_l}^{\text{ren}}$	
	APE	HYP
0.01/0.04	1.07(8)	1.10(4)
0.02/0.04	1.06(5)	1.02(3)
0.03/0.04	0.97(4)	0.99(2)

three-point correlation functions

$$\mathcal{C}_{O_i}^I(t_f, t, t_0) = a^3 \sum_{\vec{x} \in V} \langle 0 | A_0^{(+I)}(t_f)^\dagger O_i^{\text{lat}}(\vec{x}, t) A_0^{(-I)}(t_0)^\dagger | 0 \rangle, \quad (69)$$

where $A_0^{(\pm)I}(t)$ is either the box-source axial current $A_0^{(\pm)B}(t)$ or the wall-source axial current $A_0^{(\pm)W}(t)$. The operator O_i can be either the sum of the squared vector plus squared axial vector current or the squared scalar plus squared pseudoscalar current, respectively,²

$$O_{VV+AA}^{\text{lat}} = 2 \left(\bar{h}^{(+)} \gamma^\mu q \right) \left(\bar{h}^{(-)} \gamma_\mu q \right) + 2 \left(\bar{h}^{(+)} \gamma^\mu \gamma_5 q \right) \left(\bar{h}^{(-)} \gamma^\mu \gamma_5 q \right), \quad (70)$$

$$O_{SS+PP}^{\text{lat}} = 2 \left(\bar{h}^{(+)} q \right) \left(\bar{h}^{(-)} q \right) + 2 \left(\bar{h}^{(+)} \gamma_5 q \right) \left(\bar{h}^{(-)} \gamma_5 q \right). \quad (71)$$

Although the $SS + PP$ operator does not contribute to $B_q - \bar{B}_q$ mixing in continuum QCD, its counterpart in HQET is introduced through the QCD $\rightarrow$ HQET matching as shown in Eq. (27).

Because of the different spatial wavefunctions used for the APE- and HYP-smeared data, we extract the $B_q - \bar{B}_q$ matrix element in different ways for the two data sets. With the APE-smeared box source data, we can compute the matrix element directly from the ratio of correlators [57]

$$M_{O_i} = \lim_{t_f \gg t \gg t_0} 2 \frac{C_{O_i}^B(t_f, t, t_0) e^{m_{B_q}^*(t_f - t_0)/2}}{\sqrt{C^{BB}(t, t_f) C^{BB}(t, t_0)}}. \quad (72)$$

² We neglect the odd-parity parts of O_L^{lat} and O_S^{lat} , Eqs. (44) and (45), when computing the lattice three-point correlation functions because only the parity-conserving components contribute to the desired matrix element.